

Question: Lecture 5 and 6: Error Control Coding and CRC and Error Detection

A block code has $n = 10$ and $k = 8$. What is the code rate?

- A. 0.2
- B. 1.25
- C. 0.5
- D. 0.8

Answer D

A code has $d_{\min} = 5$. What is the maximum number of errors it can correct?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

If $d_{\min} = 4$, how many errors can be detected?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: C ($d_{\min} - 1$)

Why can't a code maximize both detection and correction?

- A. Bandwidth limitations
- B. Noise constraints
- C. $d_{\min}$ theorem constraints
- D. Hardware limitations

Answer: C

Which scenario best requires FEC(Forward Error Correction) ?

- A. Satellite communication
- B. WiFi
- C. Ethernet
- D. TCP transmission

Answer: A